Hangman Game Project Report

Hangman Game Using Python

A command-line word guessing game built with Python — exploring core programming concepts through interactive gameplay.

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Introduction

The Hangman Game is a simple command-line based word guessing game developed using Python. The objective of the game is for the player to correctly guess a hidden word by entering one letter at a time before all chances are exhausted.

This project was developed as part of Python fundamentals practice during the internship training program. The game helped in understanding core programming concepts such as loops, conditional statements, functions, lists, strings, and user interaction.

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Objectives

1

Python Fundamentals

To understand basic Python programming concepts

2

Problem Solving

To implement logical problem-solving techniques

3

Interactive Game

To create an interactive command-line game

4

Core Constructs

To practice functions, loops, and conditional statements

5

Code Quality

To improve debugging and code structuring skills

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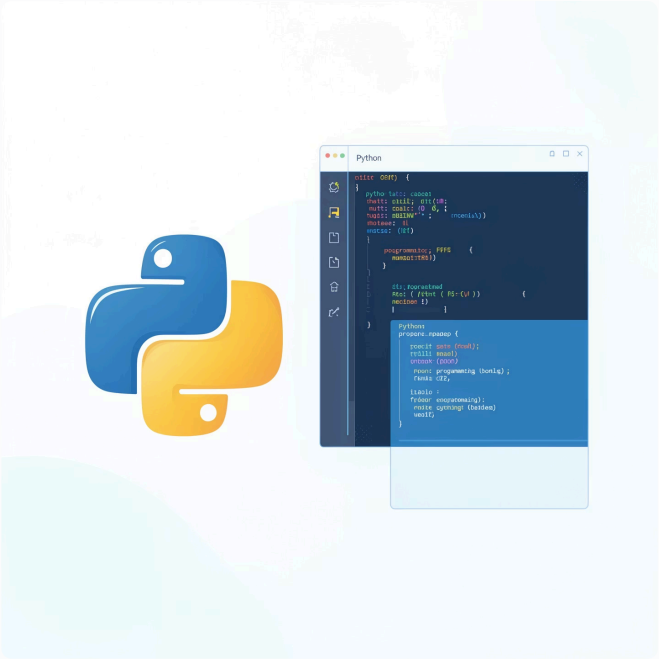
Technologies Used

Programming Language

Python

Python Concepts Used

- Lists
- Strings
- Loops
- Conditional Statements
- Functions
- Random Module
- User Input Handling



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Working of the Game

01

Word Selection

A random word is selected from a predefined list using the random module.

02

Chances Assigned

The player is given limited chances to guess the word.

03

Letter Input

The player enters one letter at a time.

04

Correct Guess

If the guessed letter exists in the word, the letter is revealed in its correct position.

05

Incorrect Guess

If the guessed letter is incorrect, the remaining chances decrease.

06

Hangman Display

A hangman figure is displayed progressively after wrong guesses.

07

Game End

The game continues until the player guesses the complete word, or all chances are exhausted.

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Features of the Game

Interactive Gameplay

User-friendly command-line interaction

Random Word Selection

Different words are selected every time the game runs

Visual Hangman Representation

ASCII-based hangman drawing displayed after incorrect guesses

Input Validation

Prevents invalid or repeated inputs

Winning & Losing Conditions

Displays appropriate messages based on result

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Modules Used

CORE MODULE

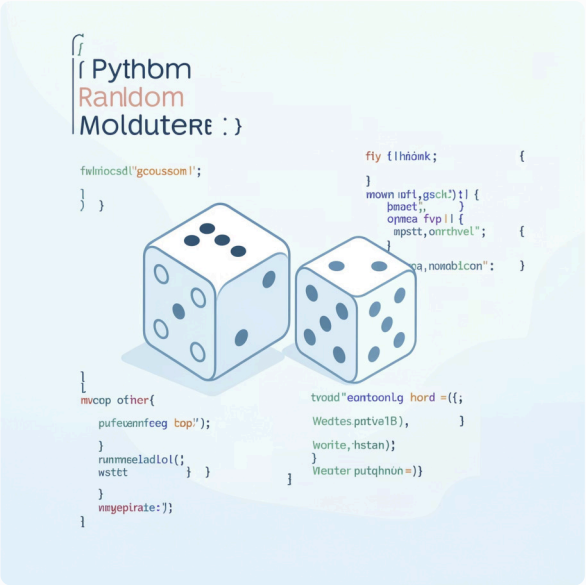
Random Module

The `random` module was used to select a random word from the predefined word list. This ensures that every game session presents a different challenge to the player.

Example usage:

```
word = random.choice(someWords)
```

> The `random.choice()` function picks a single element randomly from a list, making it ideal for word selection in games.



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Logic Used in the Program

Main Logic Components

- Random word generation
- Tracking guessed letters
- Updating hidden word display
- Counting remaining chances
- Checking win/loss conditions

Data Structures Used

List

Storing word collection

String

Word manipulation

Set / List

Tracking guessed characters

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Sample Workflow

Below is an example of how the game progresses during a typical session.

1

Selected Word

APPLE

2

Initial Display

`_ _ _ _ _`

3

User Guess: A

Updated Display: `A _ _ _ _`

4

Wrong Guess: Z

Remaining Chances: **4**

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Learning Outcomes

Through this project, the following skills were improved:

Python Syntax

Python syntax understanding

Logical Thinking

Logical thinking

Game Development

Game development basics

Functions

Function implementation

Loops & Conditions

Loop and condition handling

Debugging

Debugging techniques

User Interaction

User interaction handling

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